Introduction to genome annotation



22/09/2025

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Jacques Dainat Ph.D.

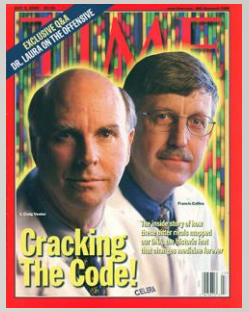


… prices go down

Human genome sequencing:

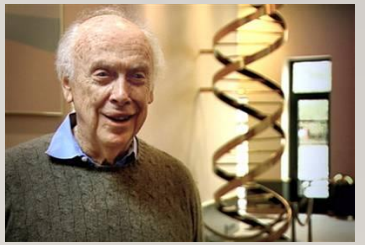
2004: Genome of Craig Wenter costs 70 mln $

- Sanger's sequencing



2007: Genome of James Watson costs 2 mln $

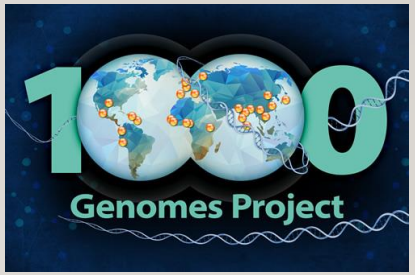
- 454 pyrosequencing



2014: Ultimate goal: 1000 $ / individual

2016: Illumina Xten: Almost there! (1200 $)

2017: NovaSeq: "Hold my beer…" (100 $)



… scientific value diminishes



Science 5 September 1997:
Vol. 277 no. 5331 pp. 1453-1462
DOI: 10.1126/science.277.5331.1453

IF 31.6

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ARTICLES

The Complete Genome Sequence of *Escherichia coli* K-12

Frederick R. Blattner*, Guy Plunkett III*, Craig A. Bloch, Nicole T. Perna, Valerie Burland, Monica Riley, Julio Collado-Vides, Jeremy D. Glasner, Christopher K. Rode, George F. Mayhew, Jason Gregor, Nelson Wayne Davis, Heather A. Kirkpatrick, Michael A. Goeden, Debra J. Rose, Bob Mau and Ying Shao

Journal of Biotechnology
Article in Press, Corrected Proof - Note to users
doi:10.1016/j.jbiotec.2010.12.018 | How to Cite or Link Using DOI
Permissions & Reprints

IF 2.9

The complete genome sequence of the dominant *Sinorhizobium meliloti* field isolate SM11 extends the *S. meliloti* pan-genome

Susanne Schneiker-Bekel[a], Daniel Wibberg[a], Thomas Bekel[b], Jochen Blom[b], Burkhard Linke[b], Heiko Neuweger[b], Michael Stiens[a, c], Frank-Jörg Vorhölter[a], Stefan Weidner[a], Alexander Goesmann[b], Alfred Pühler[a] and Andreas Schlüter[a],

perspective



*It is the **annotation** that bridges the gap from the sequence to the biology of the organism*

What is annotation ?

Structural annotation:

Find out where the regions of interest (usually genes) are in the sequence data and what they look like.

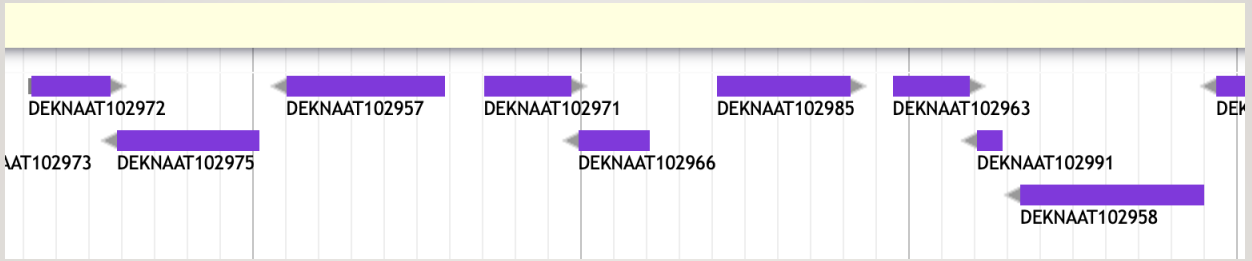
vs

functional annotation:

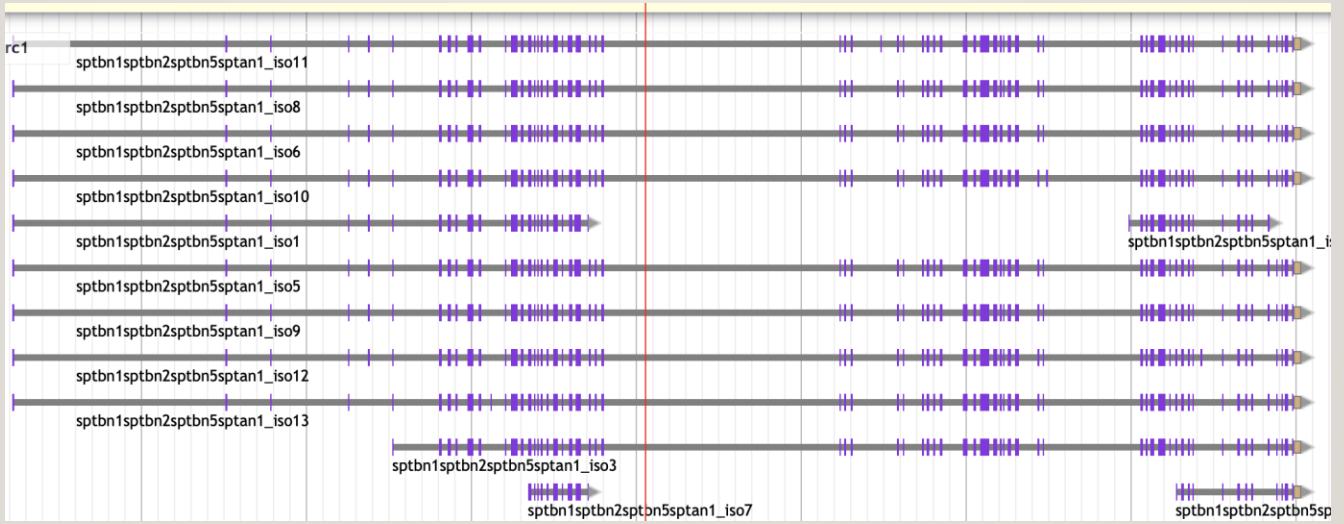
Find out what the regions do. What do they code for?

Organisms differ in genomic complexity

A yeast



A crustacean



Introduction to annotation

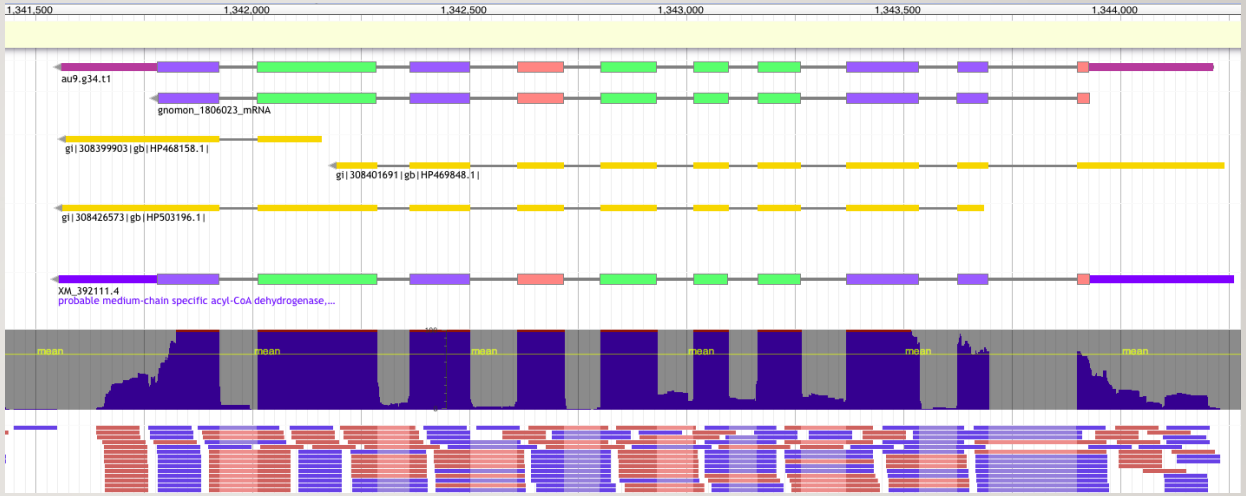
From a genome…

FASTA

```
>scaffold_26
AGTCACACACCCTTCAGCTTACACCCTGACTGCAGCCCTTACTCAAAACA
TTCCAGCCAGGAAGATGCTCCGACACAGCTTCTGGATGCCGCTCCTCGAC
GTCGAACGGCCCGCGCCGGGAAAATCGGCAGCGTCGGTGACCGCGGAGAT
CCGAAGCCGCCTCGGGGACCTGCGAGACAACGGGAGGCGGTCAACGAGAC
GCCGAGGGCTGGGAGTTATTCCCACACCGGGCCCGTAAGTTTTCTACCCA
AAAACCCATAGAAAAGAGATGAACCACTAAGTTTGATAACTCTTCTACTT
AACCGTGACCCTACGTGCCGGGGCAGGGCAGCTCTGACCCTAAGCGGCAC
ACGAACAAGGTGGTGCGCCCAATATAAACAAAGATGATGCAAGGGCTTGA
AATAAATCTCCGGAAGATTAATTCTCGAGCCCGACACGCTTTGAGGCAGC
GGAACCTACAGAACCACCGCAGTCACGTGAGAAGAGTCTAATACTCTCCA
AAGAGAAGTCCAAGGGAATGGAACGTGAAAAGAAGGTGCTTATCAAAAGC
GAGAAGGAAGATGGATGAGAACATCTTGTGTACTTCTTCTGGTCTCAAAA
AGCAAAAATGTAAAGATGCCAGACTAAGCCCGATCTGAGAAAGTACGCGA
GCAGAGACCCCCGCTGCCGATGTGGCCCAGAACGATGCCGATAAAGCACC
GAGACATAACAAAGCCCTGTGACACACAAGACGATGGACACAAACTACAT
AACACAGACACAAACTAAATGACACAGAGAGAAGTTGAAACTTCTGGGGA
AGTAAACATTTCTGAAACATCTACCAACAATCCGTCCATATATATTTCCA
TTCCACGGGACTCTTGGTTTGATATATGCGTGTTAACAGTAATCCCCGCT
GTAGCAATCACCACTATGCATAATTCATTAATTCTTTGGAGTTGCTGAGT
ATCATCTTATCAGTCTTATTTTTTTCCTTGGCTCTGGTTTCGGGCTTTTT
TTTTTTCTTCTGATAAGATTTTCCAGGAATGTGAAGACCCCCTGCATCCT
TCCCAAACTGACCACCCAAACTACAGACATTCTATAGCATTACATTACAC
AACCTAGGCAAAGTTTTTCTAACATTAAGGAACATGAAAAAAGCCAACAT
CACAATATATTCATAACAATTATGGAACATGCGAAAAGCCAATACCACAG
TACATTTATAACAATACCTCCCTTTTCCTTTCTTTAGAGATCATATGGCT
TGACCGCCGCCTCCTCGCCCGCCACCGCTGAGTACTGCCGTGCCGGAGTC
ACGGAGCCAGTCCCCCGCGGCCCCACCGCCTCCTCGCCCGCCGCCACGGA
GATCGGCTGCGCCACTCCCGAGCTCGGCCGTGCCATCGCCGCCCCCGCCG
GGGTCCCCCGGCNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNNN
```

...to an annotated gene

GTF/GFF



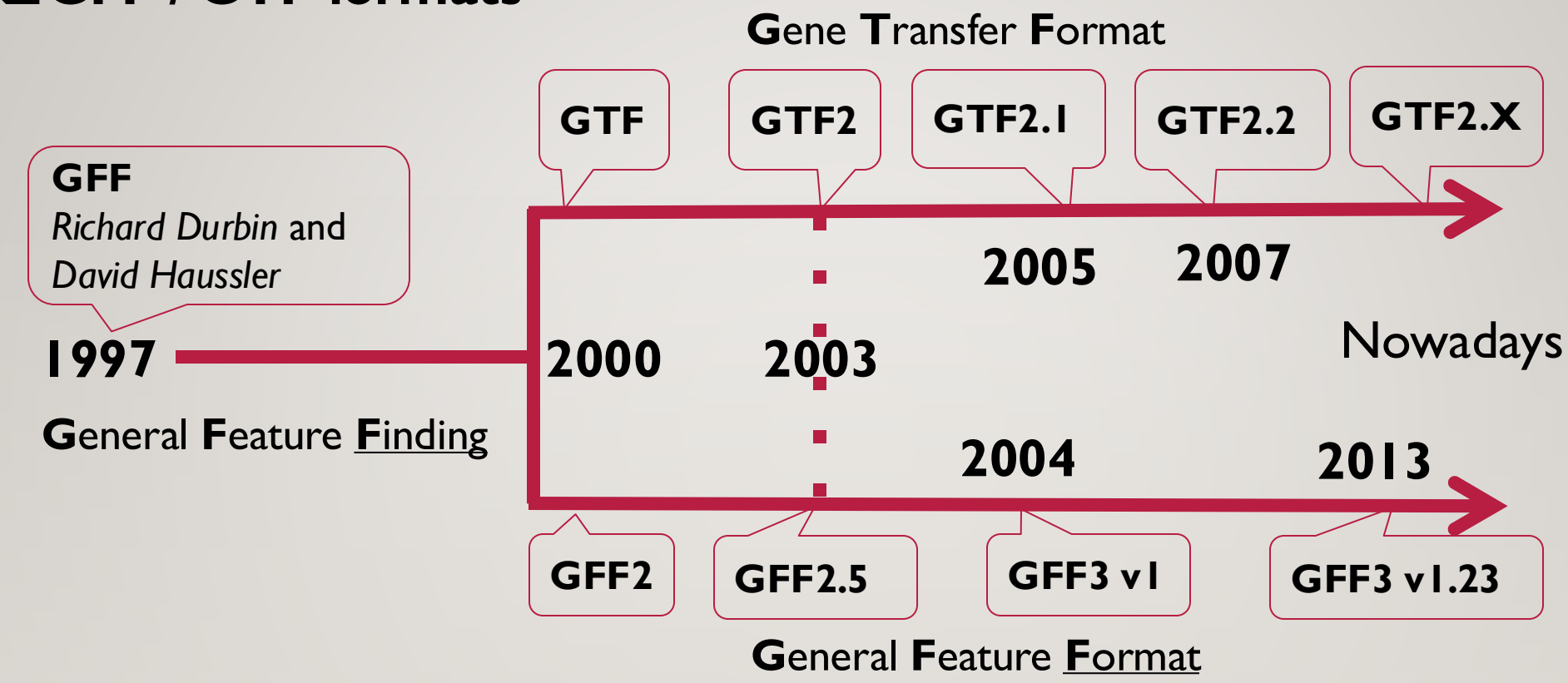
- 9 columns
- 1 feature = 1 line

2. source | 4. start | 6. score | 8. phase

Ctg123 maker Gene 1000 9000 . + . ID=gene1; Name=EDEN

1. sequence id | 3. feature type | 5. end | 7. strand | 9. attribute(s) tag=value

❑GFF / GTF formats



https://agat.readthedocs.io/en/latest/gxf.html

Introduction to annotation: GFF3

```
##gff-version 3.2.1
##sequence-region ctg123 1 1497228
ctg123  .  Gene   1000  9000  .  +  .  ID=gene1;Name=eden
ctg123  .  mRNA   1050  9000  .  +  .  ID=mRNA1;Parent=gene1;Name=eden
ctg123  .  mRNA   1050  9000  .  +  .  ID=mRNA2;Parent=gene1;Name=eden
ctg123  .  exon   1300  1500  .  +  .  ID=exon1;Parent=mRNA2;Name=eden
ctg123  .  exon   1050  1500  .  +  .  ID=exon2;Parent=mRNA1,mRNA2;Name=eden
ctg123  .  exon   3000  3902  .  +  .  ID=exon3;Parent=mRNA1;Name=eden
ctg123  .  exon   5000  5500  .  +  .  ID=exon4;Parent=mRNA1,mRNA2;Name=eden
ctg123  .  exon   7000  9000  .  +  .  ID=exon5;Parent=mRNA1,mRNA2;Name=eden
ctg123  .  CDS    1201  1500  .  +  0  ID=cds1;Parent=mRNA1;Name=eden
ctg123  .  CDS    3000  3902  .  +  0  ID=cds1;Parent=mRNA1;Name=eden
ctg123  .  CDS    5000  5500  .  +  0  ID=cds1;Parent=mRNA1;Name=eden
ctg123  .  CDS    7000  7600  .  +  0  ID=cds1;Parent=mRNA1;Name=eden
Ctg123  .  CDS    1201  1500  .  +  0  ID=cds2;Parent=mRNA2;Name=eden
ctg123  .  CDS    5000  5500  .  +  0  ID=cds2;Parent=mRNA2;Name=eden
Ctg123  .  CDS    7000  7600  .  +  0  ID=cds2;Parent=mRNA2;Name=eden
```

1 - sequence id

2 - source

3 - feature type

4 - start

5 - end

6 - score

7 - strand

8 - phase

9 - attributes
tag=value

- Header
- 9 columns
- 1 feature = 1 line
- 1 record = features with relationship

Introduction to annotation: GTF

```
#!genome-build GRCz11
#!genome-date 2017-05
ctg123  .  Gene        1000  9000  .  +  .  gene_id gene1; name eden;
ctg123  .  Transcript  1050  9000  .  +  .  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  Transcript  1050  9000  .  +  .  gene_id gene1; transcript_id=t2; name eden;
ctg123  .  exon        1300  1500  .  +  .  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  exon        1050  1500  .  +  .  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  exon        1050  1500  .  +  .  gene_id gene1; transcript_id=t2; name eden;
ctg123  .  exon        3000  3902  .  +  .  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  exon        5000  5500  .  +  .  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  exon        5000  5500  .  +  .  gene_id gene1; transcript_id=t2; name eden;
ctg123  .  exon        7000  9000  .  +  .  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  exon        7000  9000  .  +  .  gene_id gene1; transcript_id=t2; name eden;
ctg123  .  CDS         1201  1500  .  +  0  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  CDS         3000  3902  .  +  0  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  CDS         5000  5500  .  +  0  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  CDS         7000  7600  .  +  0  gene_id gene1; transcript_id=t1; name eden;
ctg123  .  CDS         1201  1500  .  +  0  gene_id gene1; transcript_id=t2; name eden;
ctg123  .  CDS         5000  5500  .  +  0  gene_id gene1; transcript_id=t2; name eden;
ctg123  .  CDS         7000  7600  .  +  0  gene_id gene1; transcript_id=t2; name eden;
```

1 - sequence id
2 - source
3 - feature type
4 - start
5 - end
6 - score
7 - strand
8 - phase
9 - attributes
tag=value

- Header
- 9 columns
- 1 feature = 1 line
- 1 record = features with relationship

GFF/GTF interoperability issues

GTF and GFF formats are similar but very different:

Different ways
- to establish the relationships between the features
- to deal with features types
- to deal with concepts (stop codon vs CDS)
- …

GFF

ctg123	.	gene	1000	9000	.	+	.	ID=gene1	Level1
ctg123	.	mRNA	1000	9000	.	+	.	ID=mRNA1;Parent=gene1	Level2
ctg123	.	exon	1000	2000	.	+	.	ID=exon1;Parent=mRNA1	Level3
ctg123	.	exon	8000	9000	.	+	.	ID=exon2;Parent=mRNA1	
ctg123	.	CDS	1700	2000	.	+	0	ID=cds01;Parent=mRNA1	
ctg123	.	CDS	8000	8300	.	+	0	ID=cds01;Parent=mRNA1	

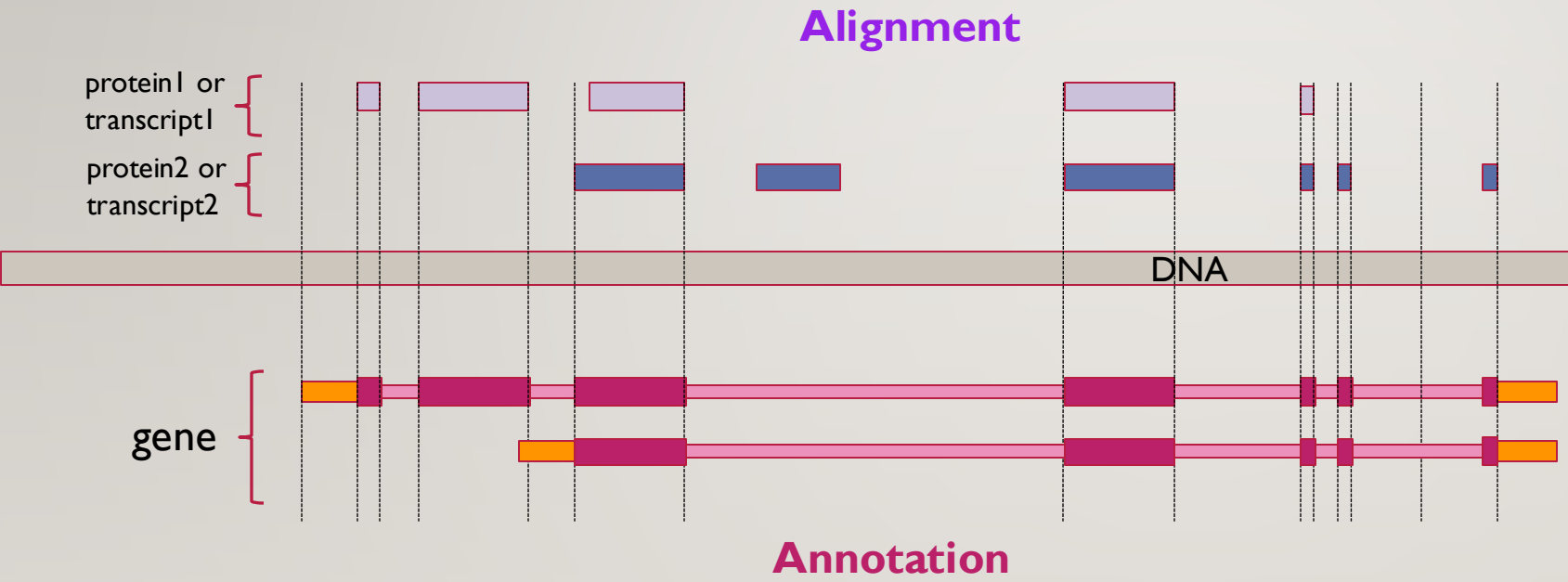
2278 terms accepted: Constrained to be either a term from the Sequence Ontology or an SO accession number

GTF

ctg123	.	gene	1000	9000	.	+	.	gene_id g1;	Level1
ctg123	.	mRNA	1000	9000	.	+	.	gene_id g1; transcript_id t1;	Level2
ctg123	.	exon	1000	2000	.	+	.	gene_id g1; transcript_id t1;	Level3
ctg123	.	exon	8000	9000	.	+	.	gene_id g1; transcript_id t1;	
ctg123	.	CDS	1700	2000	.	+	0	gene_id g1; transcript_id t1;	
ctg123	.	CDS	8000	8300	.	+	0	gene_id g1; transcript_id t1;	

9 terms accepted: gene, transcript, exon, CDS, Selenocysteine, start_codon, stop_codon, three_prime_utr and five_prime_utr

/!\ different type of gff: **alignment** / **annotation** / other



match, match_part

Gene, mRNA, exon, intron, CDS, splice site, UTR, … isoforms

The main steps in genome annotation

1 2 3 4 5

QC assembly → Structural annotation → Manual curation → Functional annotation → Downstream analysis / Submission

BUSCO

MAKER Annotate this!

Web Apollo

InterPro



QC assembly

The quality of the assembly will heavily influence the quality of the annotation

- ❑SNP-errors can change start/stop-codons/splice sites
- ❑ Indels can cause frame-shifts
- ❑High fragmentation could break loci
- ❑missing loci cannot be annotated

=> Annotation tools have difficulties to deal with those problems

Before to start a project check:

- ❑ Fragmentation (N50, number of sequences, how many small contigs)
- ❑ Sanity of the fasta file (Ns, IUPAC, lowercase nucleotides)
- ❑ Completeness / duplication / fragmentation
- ❑ **Presence of Organelles**
- ❑ Other (GC content, how distant from other species)



BUSCO version is: 3.0.2
The lineage dataset is: **fungi_odb9** (Creation date: 2016-02-13, number of species: 85, number of BUSCOs: **290**)
#
Summarized benchmarking in BUSCO notation for file genome.fa
BUSCO was run in mode: **genome**

C:98.6%[S:97.9%,D:0.7%],F:0.0%,M:1.4%,n:290

286	Complete BUSCOs (C)
284	Complete and single-copy BUSCOs (S)
2	Complete and duplicated BUSCOs (D)
0	Fragmented BUSCOs (F)
4	**Missing BUSCOs (M)**
290	Total BUSCO groups searched

What to annotate?

Feature type	DB associated	Tool example
Coding genes	Ensembl, NCBI, EBI, Uniprot	https://juke34.github.io/awesome-genome-annotation/
ncRNA	Rfam	infernal
tRNA	Sprinzl database	tRNAscan-SE
snoRNA		snoscan
miRNA	miRBase	Splign miR-PREFeR (for plant)
Repeats	Repbase, Dfam	RepeatModeler, repeatMasker, REPET, EDTA
Pseudogenes		pseudopipe
…		

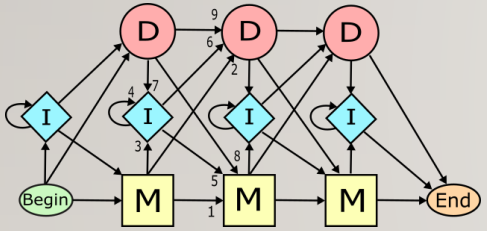
=> Focus on Repeats and coding genes

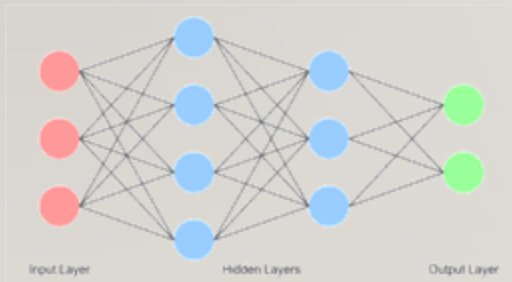
Why repeats?

Types of data needed

Machine Learning

- HMM
- Deep Learning





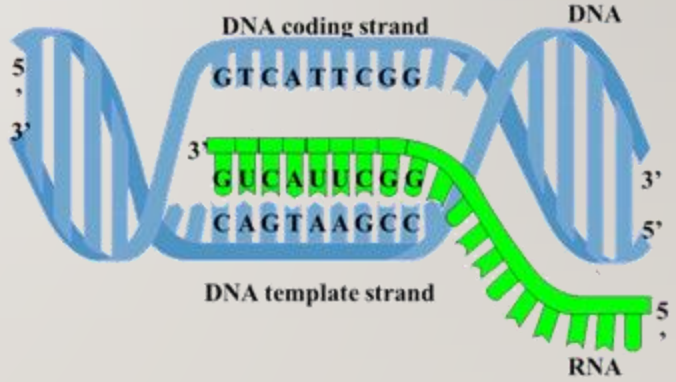
Proteins

- Known amino acid sequences from other organisms



Transcripts

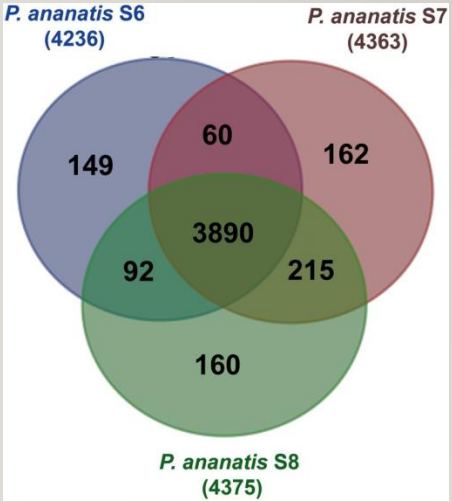
- Assembled from RNA-seq or downloaded ESTs



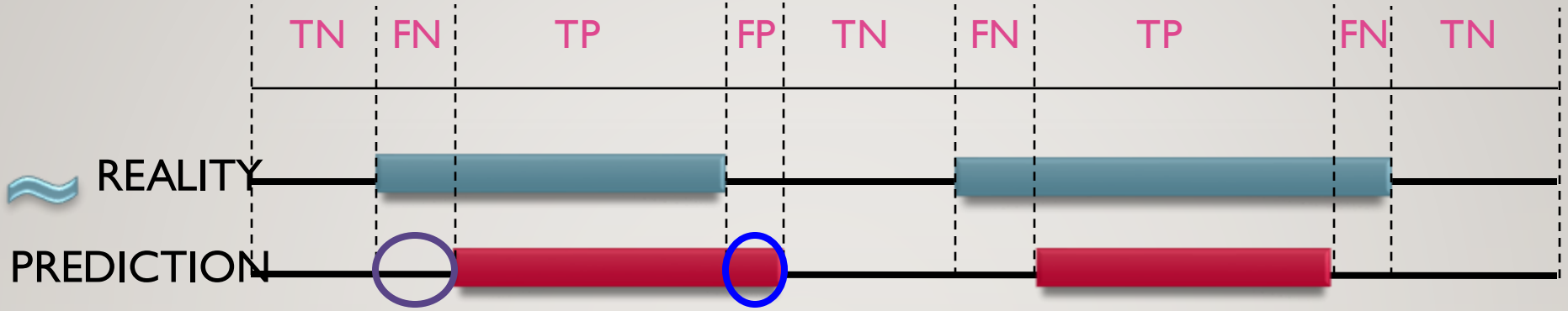
Assessing an annotation

No annotation method is perfect, they do mistakes !!

- ❑ Simple statistics (number genes / number exon per gene)
- ❑ BUSCO (and compare against assembly result)
- ❑ Protein/transcript evidence (AED score in MAKER)
- ❑ Comparative genomics
- ❑ Domain / Function attached
- ❑ Visualization



Assess the quality of the *ab-initio* model/training:



Sensitivity is the proportion of true predictions compared to the total number of correct genes (including missed predictions)

$$Sn = \frac{TP}{TP + FN}$$

Specificity is the proportion of true predictions among all predicted genes (including incorrectly predicted ones)

$$Sp = \frac{TP}{TP + FP}$$

Ab Initio methods can approach 100% sensitivity, however as the sensitivity increases, accuracy suffers as a result of increased false positives.

Assessing an annotation

```
*******      Evaluation of gene prediction     *******

-----------------------------------------\
                | sensitivity | specificity |
-----------------------------------------|
nucleotide level |       0.987 |       0.896 |
-----------------------------------------/

-----------------------------------------------------------------------------------------------------\
           |  #pred |  #anno |      |    FP = false pos.  |    FN = false neg.  |             |             |
           | total/ | total/ |   TP |---------------------|---------------------| sensitivity | specificity |
           | unique | unique |      | part | ovlp | wrng | part | ovlp | wrng |             |             |
-----------------------------------------------------------------------------------------------------|
           |        |        |      |                 85 |                  45 |             |             |
exon level |    512 |    472 |  427 | ------------------ | ------------------ |       0.905 |       0.834 |
           |    512 |    472 |      |   29 |    2 |   54 |   30 |    1 |   14 |             |             |
-----------------------------------------------------------------------------------------------------/

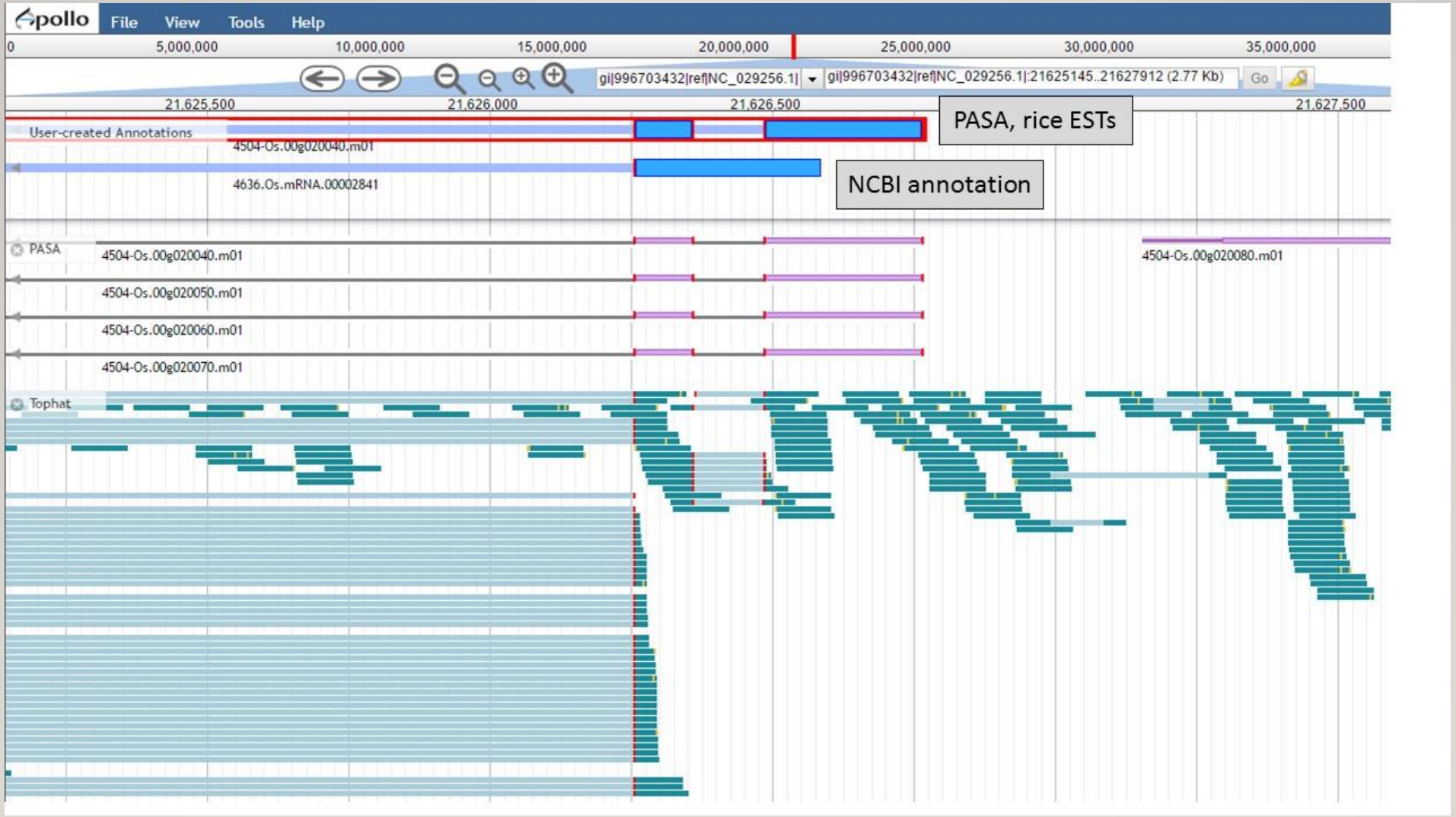
-----------------------------------------------------------------------\
transcript | #pred | #anno |   TP |   FP |   FN | sensitivity | specificity |
-----------------------------------------------------------------------|
gene level |   105 |   100 |   67 |   38 |   33 |        0.67 |       0.638 |
-----------------------------------------------------------------------/
```

Visualization / Manual curation tools

Name	Standalone	Web tool	Manual curation	year	comment
Artemis	X		X	2000	Can save annotation in EMBL format
IGV	X			2011	Popular
Savant	X			2010	Sequence Annotation,Visualization and ANalysis Tool. enable Plug-ins
Tablet	X		X	2013	
IGB	X			2008	enable Plug-ins. Can load local and remote data (dropbox, UCSC genome, etc)
Jbrowse		X		2010	GMOD (successor of Gbrowse)
Web Apollo		X	X	2013	Active community (gmod). Based on Jbrowse. Real-time collaboration
UCSC		X		2000	A large amount of locally stored data must be uploaded to servers across the internet
Ensembl genome browsers		X		2002	A large amount of locally stored data must be uploaded to servers across the internet

Annotation can be improved by **manual curation**

Visualization / Manual curation tools



Functional annotation

Most methods only build gene models, no **functional inference**

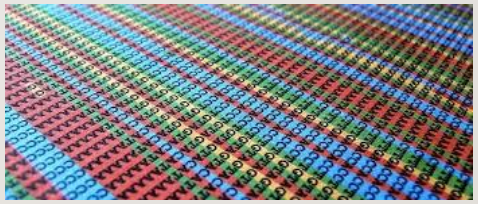
Accurate

- Experimentally
  => Mutants, knockout, etc.



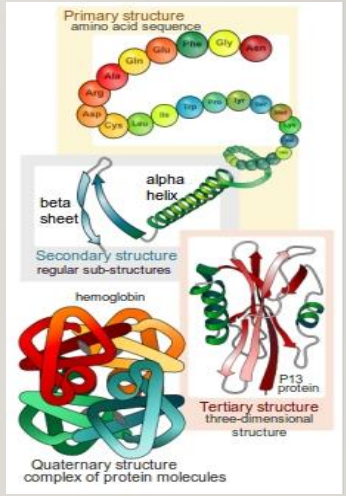
Mice homozygous for the diabetes 3J spontaneous mutation

- Computationally
  - Sequence-based

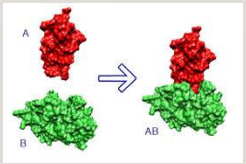


limited accuracy

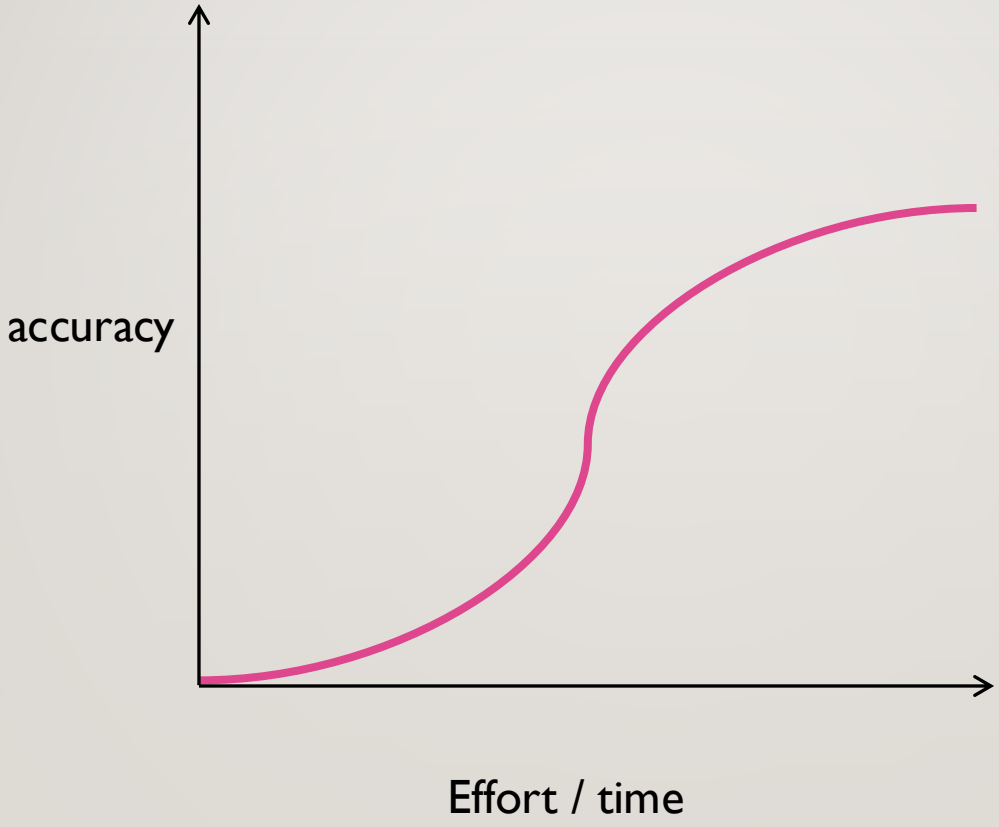
  - Structure based



  - Protein-protein interaction data



Effort versus accuracy

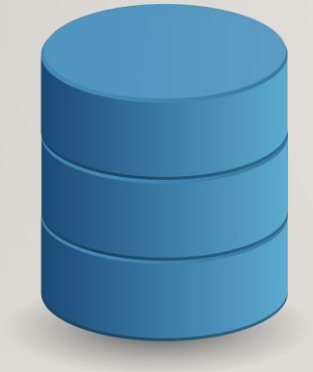


Data Submission

Submit your annotation in public archives e.g. EBI, NCBI, etc.

GFF





	EBI	NCBI
converter	EMBLmyGFF3	GAG Manual + table2asn
Expected format	Flat file (EMBL)	ASN.1 format


The End
Thank you for your attention